

1. Some icy cold liquid is kept cold inside a vacuum flask. Which statements are correct ?
- (1) The flask's cork stopper reduces heat gain from the surroundings.
  - (2) The silver coating on the inner surface of the glass wall is a good reflector of infra-red.
  - (3) The vacuum between the double glass walls reduces heat gain by radiation.
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

2. 0.3 kg of water at temperature  $50^{\circ}\text{C}$  is mixed with 0.2 kg of ice at temperature  $0^{\circ}\text{C}$  in an insulated container of negligible heat capacity. What is the final temperature of the mixture ?

Given: specific heat capacity of water  $= 4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

specific latent heat of fusion of ice  $= 3.34 \times 10^5 \text{ J kg}^{-1}$

- A.  $-1.8^{\circ}\text{C}$
- B.  $0^{\circ}\text{C}$
- C.  $1.8^{\circ}\text{C}$
- D.  $3.0^{\circ}\text{C}$

\*3. When an ideal gas is heated from 25 °C to 50 °C, the average kinetic energy of the gas molecules will

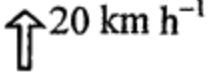
- A. double.
- B. increase by 41 %.
- C. increase by 8.4%.
- D. increase by 4.1%.

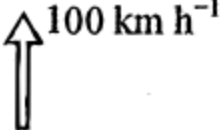
4. The speedometer of a car shown below indicates the car's



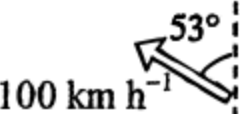
- A. instantaneous speed.
- B. instantaneous velocity.
- C. average speed of the whole journey.
- D. average velocity of the whole journey.

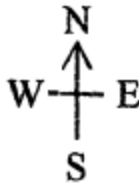
5. A car travelling at  $80 \text{ km h}^{-1}$  due east changes direction and travels at  $60 \text{ km h}^{-1}$  due north. Which diagram represents the change in velocity of the car ?

A.   $20 \text{ km h}^{-1}$

B.   $100 \text{ km h}^{-1}$

C.   $100 \text{ km h}^{-1}$

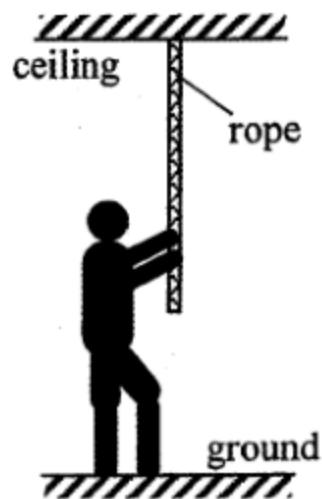
D.   $100 \text{ km h}^{-1}$



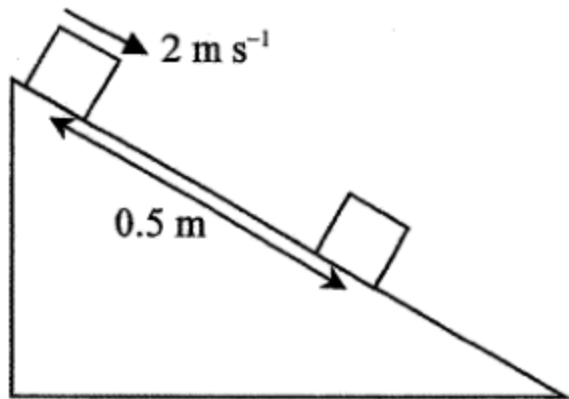
6. A boy of weight  $W$  exerts a downward pulling force  $F$  on a rope of weight  $G$  hung vertically from the ceiling. He stands still on the ground as shown. Which of the following gives the magnitude of the force exerted by

- (1) the boy on the ground;  
(2) the rope on the ceiling ?

|    | (1)     | (2)     |
|----|---------|---------|
| A. | $W$     | $G - F$ |
| B. | $W$     | $G + F$ |
| C. | $W - F$ | $G - F$ |
| D. | $W - F$ | $G + F$ |

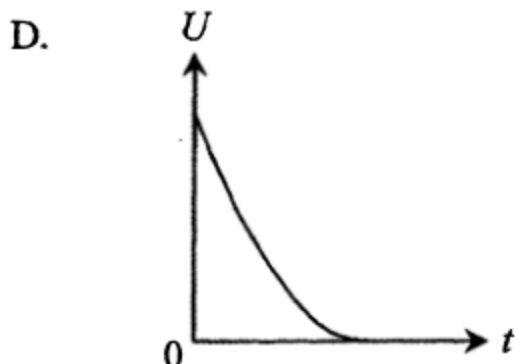
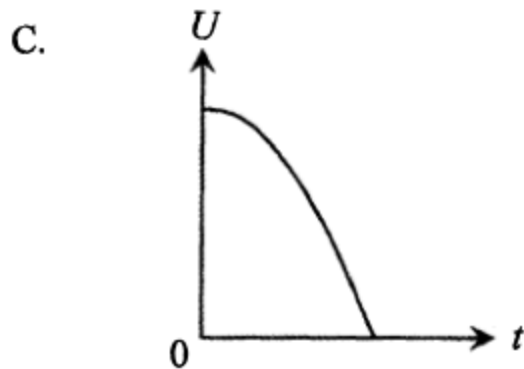
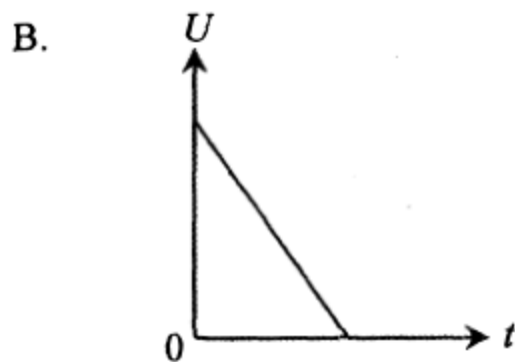
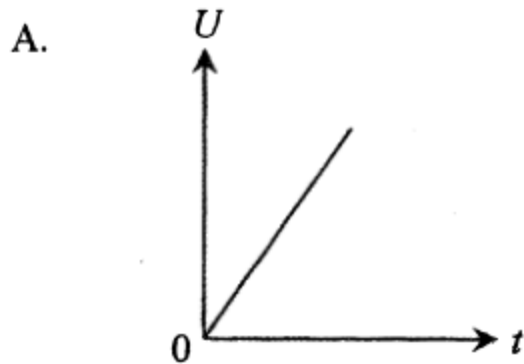


7. A block with initial speed  $2 \text{ m s}^{-1}$  slides down a rough inclined plane and stops after travelling a distance of  $0.5 \text{ m}$ . What is the deceleration of the block?



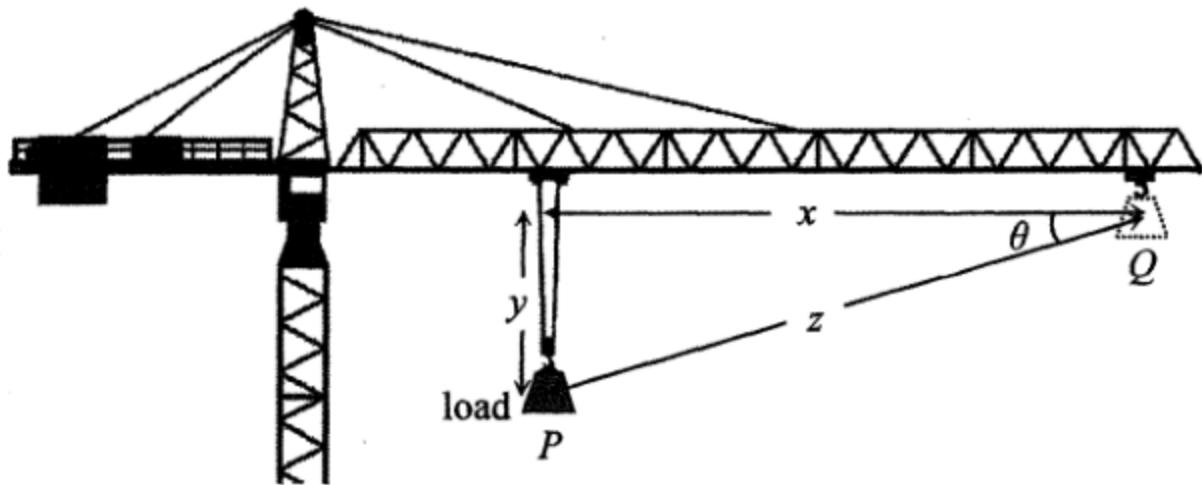
- A.  $1 \text{ m s}^{-2}$   
B.  $2 \text{ m s}^{-2}$   
C.  $4 \text{ m s}^{-2}$   
D. Answer cannot be found as the angle of inclination of the plane is not given.

8. An object at a certain height falls freely from rest under gravity. Which graph correctly shows the variation of its gravitational potential energy  $U$  with time  $t$ ? Neglect air resistance and take  $U = 0$  at the ground.





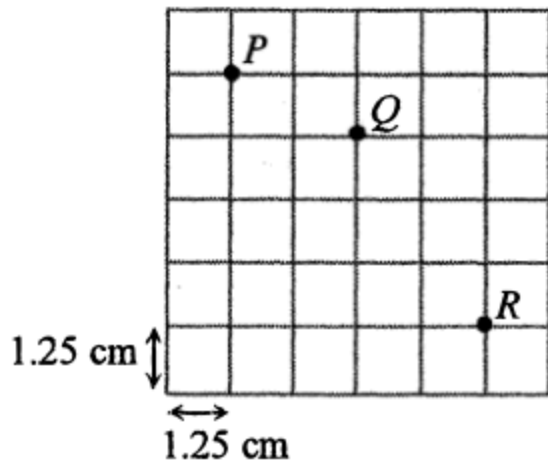
9. A crane moves a load of weight  $W$  steadily from point  $P$  to point  $Q$  as shown.



The work done on the load by the crane is

- A.  $Wy$ .
- B.  $W(x + y)$ .
- C.  $Wz$ .
- D.  $Wz \cos \theta$ .

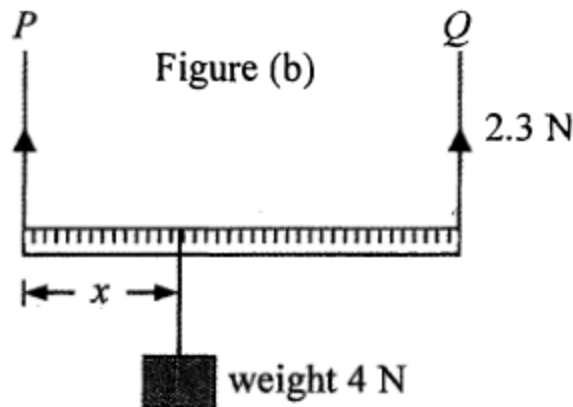
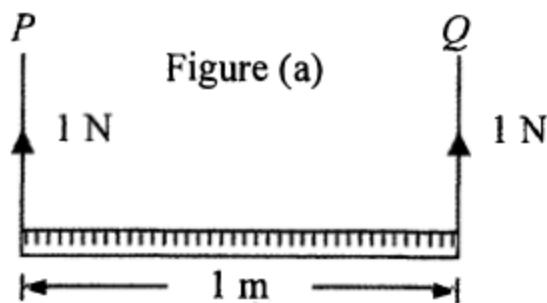
\*10.



The above stroboscopic picture shows a particle projected horizontally at position  $P$  into the air in a vertical plane. Subsequently the particle reaches positions  $Q$  and  $R$  such that the time interval between  $P$  and  $Q$  is equal to that between  $Q$  and  $R$ . Each square of the grid measures  $1.25 \text{ cm} \times 1.25 \text{ cm}$ . Find the particle's speed of projection at  $P$ . Neglect air resistance. ( $g = 9.81 \text{ m s}^{-2}$ )

- A.  $0.3 \text{ m s}^{-1}$
- B.  $0.4 \text{ m s}^{-1}$
- C.  $0.5 \text{ m s}^{-1}$
- D.  $0.6 \text{ m s}^{-1}$

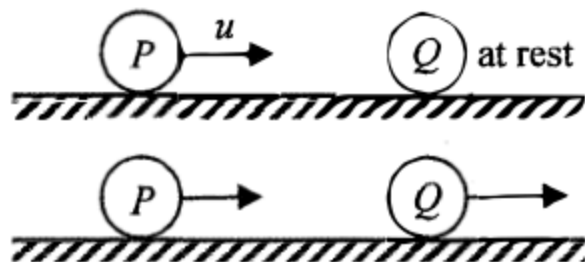
11.



A uniform metre rule is supported by vertical wires  $P$  and  $Q$  and remains at rest horizontally as shown in Figure (a). The tension in each wire is  $1\text{ N}$ . When a weight of  $4\text{ N}$  is hung from the metre rule at a certain position as shown in Figure (b), the tension in  $Q$  becomes  $2.3\text{ N}$  while the metre rule remains horizontal. Find the distance  $x$  shown.

- A.  $32.5\text{ cm}$
- B.  $57.5\text{ cm}$
- C.  $67.5\text{ cm}$
- D. Answer cannot be found as the tension in  $P$  is not known.

12. On a smooth horizontal surface, a marble  $P$  moving with speed  $u$  collides head-on with another marble  $Q$ , which is at rest. After collision,  $P$  and  $Q$  move with different speeds as shown.

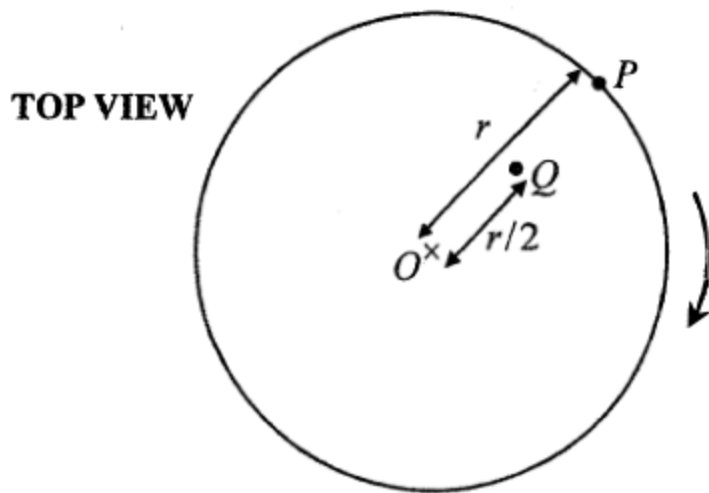


Which of the following statements about this collision is/are correct ?

- (1) During collision, the force acting on  $Q$  by  $P$  is equal and opposite to that acting on  $P$  by  $Q$ .
- (2) The total momentum of the two marbles is conserved only when the collision is perfectly elastic.
- (3) The kinetic energy lost by  $P$  must be equal to that gained by  $Q$ .

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

- \*13. Particles  $P$  and  $Q$  are fixed at distances  $r$  and  $r/2$  respectively from the centre  $O$  of a horizontal circular platform which is rotating uniformly as shown.



The ratio of the acceleration of  $P$  to that of  $Q$  is

- A. 1 : 2.
- B. 2 : 1.
- C. 1 : 4.
- D. 4 : 1.

\*14. A satellite orbits the Earth in a circular path of radius  $7.2 \times 10^6$  m. What is the period of the satellite ?

Given : mass of the Earth =  $6.0 \times 10^{24}$  kg

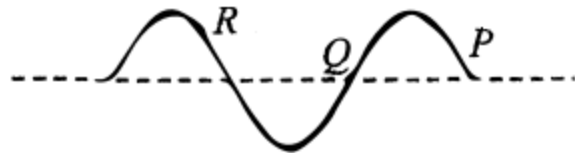
A. 1.4 hours

B. 1.7 hours

C. 1 day

D. Answer cannot be found as the mass of the satellite is not known.

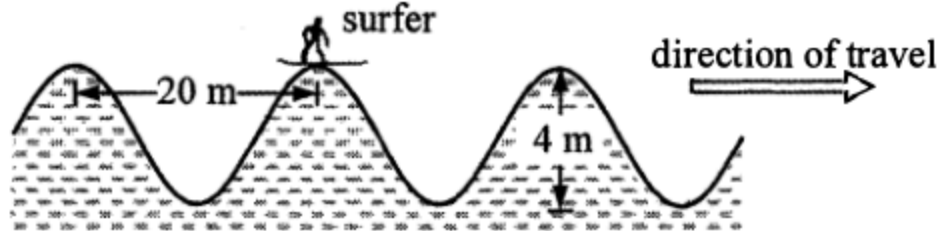
15.



The above figure shows a snapshot of a transverse wave which travels along a string. Which statement is correct ?

- A. The wave is travelling to the left if particle  $P$  is moving upwards at this instant.
- B. Particles  $P$  and  $R$  are moving in the same direction at this instant.
- C. Particle  $Q$  is at rest at this instant.
- D. Particle  $R$  vibrates with an amplitude larger than that of particle  $Q$ .

16.

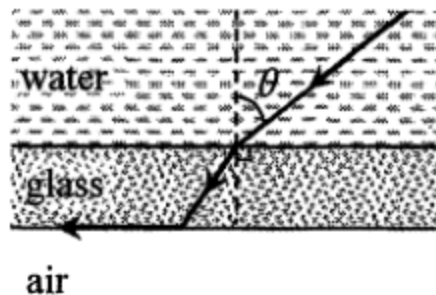


The surfer in the figure reaches a crest at the moment shown. The crests of the water wave are 20 m apart and the surfer descends a vertical distance of 4 m from a crest to a trough in a time interval of 2 s. What is the speed of the wave ?

- A.  $1 \text{ m s}^{-1}$
- B.  $2 \text{ m s}^{-1}$
- C.  $5 \text{ m s}^{-1}$
- D.  $10 \text{ m s}^{-1}$



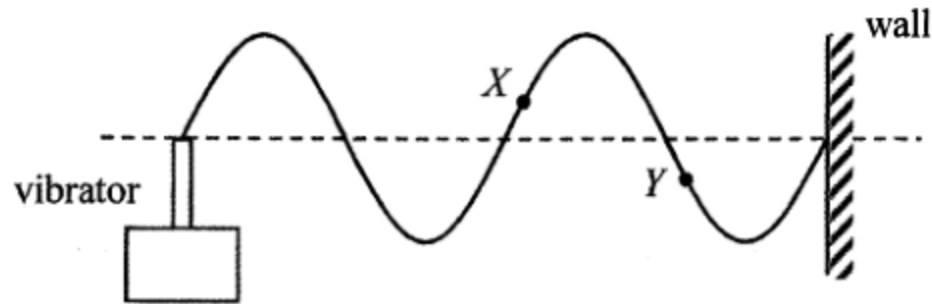
17.



A parallel-sided glass sheet separates water from air. A ray of light in water is incident at an angle  $\theta$  on the glass sheet and finally emerges into air along the glass-air interface as shown. Find  $\theta$ .  
Given: refractive index of water is 1.33.

- A.  $41.2^\circ$
- B.  $48.8^\circ$
- C.  $53.1^\circ$
- D. It depends on the refractive index of glass.

18. A string is tied to a vibrator while the other end is fixed to a wall. A stationary wave is formed as shown.



Which statement is correct when the frequency of the vibrator doubles ?

- A. The wavelength will double.
- B. The wave speed will double.
- C. The amplitude will be halved.
- D. Particles *X* and *Y* will become vibrating in phase.

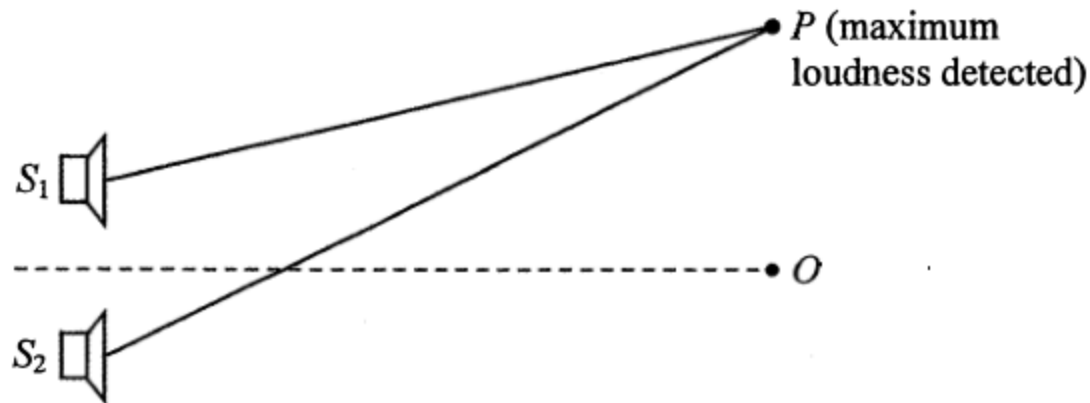
19. Diffraction will occur when light

- (1) passes through a pinhole.
- (2) passes by a sharp edge.
- (3) passes through a slit.

- A. (1) only
- B. (2) only
- C. (3) only
- D. (1), (2) and (3)

20. A beam of white light is separated into different colours after entering a glass prism because lights of different colours

- A. are diffracted to different extents by the prism.
- B. undergo total internal reflection at different angles inside the prism.
- C. travel at different speeds in vacuum.
- D. travel at different speeds in glass.



Loudspeakers  $S_1$  and  $S_2$  connected to a signal generator emit sound waves which are in phase. Point  $O$  is equidistant from the loudspeakers while at point  $P$  maximum loudness is detected. The wavelength of the sound waves is  $\lambda$ . Which statement is **INCORRECT** ?

- A. Both  $PS_1$  and  $PS_2$  must be integral multiples of wavelength  $\lambda$ .
- B. The definite value of the path difference  $PS_2 - PS_1$  cannot be determined from the information given.
- C. At least one point of minimum loudness can be detected between  $O$  and  $P$ .
- D. Minimum loudness will be detected at  $P$  if the sound waves from  $S_1$  and  $S_2$  are in antiphase.

22. An object is moving at constant speed towards a convex lens of focal length 10 cm. At the moment when it is at 100 cm from the lens, which of the following descriptions of the image is correct ?

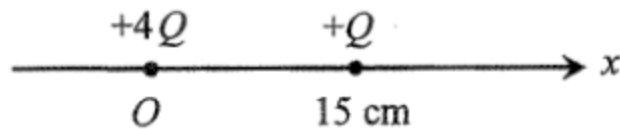
|    | direction of image movement | speed of the image             |
|----|-----------------------------|--------------------------------|
| A. | away from the lens          | faster than that of the object |
| B. | towards the lens            | faster than that of the object |
| C. | away from the lens          | slower than that of the object |
| D. | towards the lens            | slower than that of the object |

23. Which of the following are applications of ultrasound ?

- (1) sterilizing drinking water
- (2) detecting cracks in railway tracks
- (3) breaking up kidney stones

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

\*24.

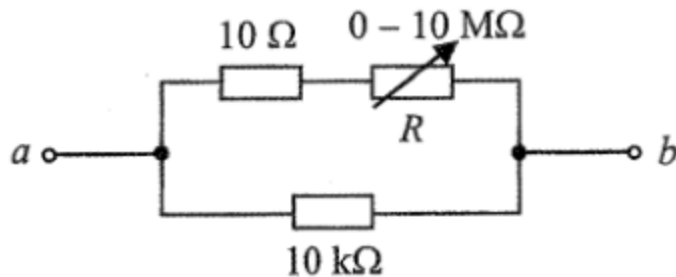


Point charges  $+4Q$  and  $+Q$  are fixed on the  $x$ -axis with  $+4Q$  at the origin  $O$  and  $+Q$  at  $x = 15\text{ cm}$  as shown. The respective electric fields due to the two charges are equal at

- A.  $x = 10\text{ cm}$ .
- B.  $x = 12\text{ cm}$ .
- C.  $x = 20\text{ cm}$ .
- D.  $x = 30\text{ cm}$ .



25.



In the above circuit, the variable resistor  $R$  can be adjusted over its full range from 0 to  $10\ \text{M}\Omega$ . What is the approximate range of resistance between  $a$  and  $b$ ?

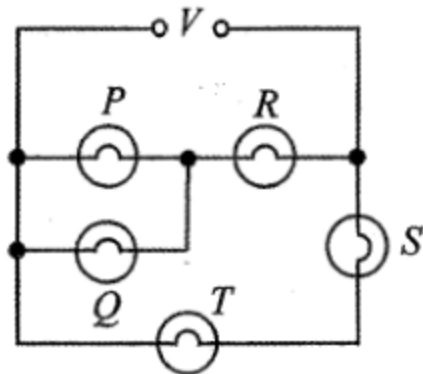
- A. 0 to  $10\ \text{k}\Omega$
- B.  $10\ \Omega$  to  $10\ \text{k}\Omega$
- C.  $10\ \Omega$  to  $10\ \text{M}\Omega$
- D.  $10\ \text{k}\Omega$  to  $10\ \text{M}\Omega$

26. Two filament light bulbs  $X$  and  $Y$  are connected in parallel to a dry cell.  $X$  is brighter than  $Y$ . Which statements are correct ?

- (1) In 1 s, the number of charges flowing through  $X$  is greater than that flowing through  $Y$ .
- (2) In 1 s, the electrical energy dissipated by  $X$  is greater than that dissipated by  $Y$ .
- (3) For every unit charge passing, the electrical energy dissipated by  $X$  is equal to that dissipated by  $Y$ .

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

27.



In the above circuit, all the bulbs are identical. If the voltage  $V$  gradually increases, which bulb(s) will burn out first?

- A.  $P$  and  $Q$
- B.  $R$
- C.  $S$
- D.  $T$

28. A television set in stand-by mode consumes 1.5 W. If it is in this mode for 16 hours a day, estimate the carbon dioxide ( $\text{CO}_2$ ) emission due to the electricity consumed in stand-by mode in a 30-day month. Given: 1 kWh of electricity consumed corresponds to 0.8 kg  $\text{CO}_2$  emission from the power station.

- A. 0.576 kg
- B. 0.720 kg
- C. 576 kg
- D. 720 kg

\*29. A student uses a search coil to study the strength of the magnetic field inside a long solenoid which is connected to an a.c. signal generator set at a certain frequency. Which of the following can improve the accuracy of this experiment ?

- (1) Ensure the plane of the search coil is perpendicular to the field lines.
- (2) Increase the signal generator's frequency and use the same current as before.
- (3) Set the axis of the solenoid along an east-west direction to avoid the effects of the Earth's magnetic field.

- A. (1) only
- B. (1) and (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

\*30. A sinusoidal a.c. of a certain frequency delivers a r.m.s. voltage  $V_{\text{r.m.s.}}$ . If its frequency is doubled and its peak voltage is halved, what would be the r.m.s. voltage ?

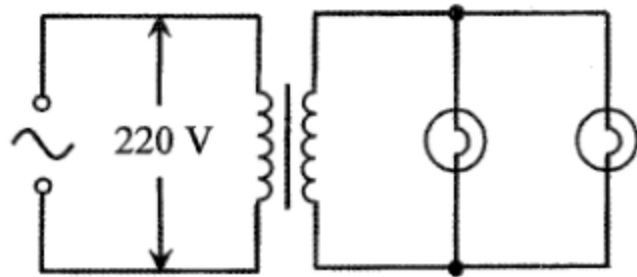
A.  $\frac{1}{2} V_{\text{r.m.s.}}$

B.  $\frac{1}{\sqrt{2}} V_{\text{r.m.s.}}$

C.  $\frac{1}{2\sqrt{2}} V_{\text{r.m.s.}}$

D.  $V_{\text{r.m.s.}}$

\*31.



In the above circuit, each light bulb works at its rated value '22 W, 11 V'. The current in the primary coil is 0.25 A. Find the efficiency of the transformer.

- A. 20%
- B. 40%
- C. 64%
- D. 80%

32. Which of the following statements about ionizing radiations is/are correct ?

- (1) The ionizing power of  $\alpha$ -particles is much stronger than that of  $\beta$ -particles.
- (2)  $\gamma$ -radiation can be completely shielded by a 10 cm thick concrete wall.
- (3) Ionizing radiations  $\alpha$ ,  $\beta$  and  $\gamma$  all undergo deflection in an electric field.

- A. (1) only
- B. (1) and (2) only
- C. (1) and (3) only
- D. (2) and (3) only



33. Two radionuclides  $X$  and  $Y$  are of half-lives 3 hours and 4 hours respectively and initially there are  $N_X$  and  $N_Y$  undecayed nuclei respectively. After 24 hours, the number of undecayed nuclei of both nuclides becomes the same. Find  $N_X : N_Y$ .

- A. 8 : 1
- B. 4 : 3
- C. 4 : 1
- D. 2 : 1

**END OF SECTION A**